



(USA) Senior Manager, Data Engineering

[Apply](#) Sunnyvale, CA Full time Posted 11 Days Ago R-2404730

[Position Summary...](#)

[What you'll do...](#)

The Mission

At Sam's Club, we are no longer just building dashboards or static pipelines; we are building the "brain" of the retail experience. As the Senior Engineering Manager for Agentic Data, you will be at the forefront of the AI revolution. Your mission is to evolve our massive data ecosystem into a Semantic & Contextual Layer—the foundational intelligence that allows AI Agents to reason, plan, and act autonomously for millions of members.

You aren't just managing a team; you are architecting the bridge between raw data and autonomous action. You will lead the transition from traditional data structures to agent-ready environments, where data is not just stored, but understood in real-time.

What You'll Do

- Architect the Semantic Future: Define the technical vision for a unified semantic layer that transforms structured, semi-structured, and unstructured data into context-aware representations (Embeddings, Knowledge

Graphs, and Metadata) consumable by LLMs and multi-agent workflows.

- **Real-Time Agentic Intelligence:** Own the streaming architecture that feeds our agents. You will leverage Kafka to build low-latency event-driven pipelines, ensuring agents have access to "up-to-the-second" member context and environmental state.
- **Champion Data Fundamentals:** Ensure the bedrock of our AI strategy is built on elite Data Engineering principles. You will oversee the design of scalable schemas, partitioning strategies, and high-performance indexing that make petabyte-scale data accessible to reasoning models.
- **Build Agent-Ready Ecosystems:** Lead the design of "Agentic Data" pipelines—ensuring that data provided to AI agents is high-fidelity, optimized for retrieval, and enriched with the necessary business logic to prevent hallucinations.
- **Strategic Leadership:** Manage and mentor a high-performing team of engineers. You will bridge the gap between AI/ML Research, Product, and Platform Engineering to align data roadmaps with long-term autonomous decision-making goals.
- **Innovate Retrieval Systems:** Own the strategy for Retrieval-Augmented Generation (RAG) at scale, optimizing how our agents discover and interact with Sam's Club's vast proprietary knowledge base across both batch and streaming sources.
- **Establish Operational Excellence:** Define standards for "Agentic Observability"—implementing telemetry, lineage, and auditability patterns specifically for autonomous and regulated AI workflows.

What You'll Bring

- **Experience:** 7–10+ years in Data Engineering, building and operating large-scale, distributed, fault-tolerant data platforms, with at least 3+ years in a leadership role.
- **Data Engineering Mastery:** Deep-rooted expertise in Data Fundamentals. You must be an expert in data modeling (Star/Snowflake, Data Vault), storage formats (Parquet, Avro), and distributed computing concepts (sharding, replication, and CAP theorem).
- **Real-Time Expertise:** Deep experience with Kafka and event-driven architectures. You understand how to manage state, schema evolution, and consistency in streaming environments (Kafka Streams, Flink, or Spark Streaming).
- **Semantic & AI Mastery:** Deep familiarity with the modern AI stack—specifically Vector Databases (Pinecone, Milvus), Knowledge Graphs, and framework orchestration (LangChain, LlamaIndex, CrewAI).
- **Cloud Scale Mastery:** Proven track record in GCP or Azure, utilizing BigQuery, Dataflow, and Pub/Sub to handle petabyte-scale workloads.
- **Data Modeling for LLMs:** Expert-level understanding of how schema choices, latency, and context window management impact AI reasoning and retrieval quality.
- **Engineering Rigor:** Fluency in Python, Java, or Scala, with a deep engineering mindset around testability, maintainability, and high-performance system design.

Benefits & Perks:

Beyond competitive pay, you can receive incentive awards for your performance. Other great perks include 401(k) match, stock purchase plan, paid maternity and parental leave, PTO, multiple health plans, and much more.

Equal Opportunity Employer:

We believe we are best equipped to help our associates, customers and the communities we serve live better when we really know them. That means understanding, respecting and valuing diversity- unique styles, experiences, identities, ideas and opinions – while being inclusive of all people.

About Global Tech

Imagine working in an environment where one line of code can make life easier for hundreds of millions of people and put a smile on their face. That's what we do at Walmart Global Tech. We're a team of 15,000+ software engineers, data scientists and service professionals within Walmart, the world's largest retailer, delivering innovations that improve how our customers shop and empower our 2.2 million associates. To others, innovation looks like an app, service or some code, but Walmart has always been about people. People are why we innovate, and people power our innovations. *Being human-led is our true disruption.*

We're virtual

Working virtually this year has helped us make quicker decisions, remove location barriers across our global team, be more flexible in our personal lives and spend less time commuting. Today, we are reimagining the tech workplace of the future by making a permanent transition to virtual work for most of our team. Of course, being together in person is an important part of our culture and shared success. We'll collaborate in person at a regular cadence and with purpose.

At Walmart, we offer competitive pay as well as performance-based bonus awards and other great benefits for a happier mind, body, and wallet. Health benefits include medical, vision and dental coverage. Financial benefits include 401(k), stock purchase and company-paid life insurance. Paid time off benefits include PTO (including sick leave), parental leave, family care leave, bereavement, jury duty, and voting. Other benefits include short-term and long-term disability, company discounts, Military Leave Pay, adoption and surrogacy expense reimbursement, and more.

You will also receive PTO and/or PPTO that can be used for vacation, sick leave, holidays, or other purposes. The amount you receive depends on your job classification and length of employment. It will meet or exceed the requirements of paid sick leave laws, where applicable.

For information about PTO, see <https://one.walmart.com/notices>.

Live Better U is a Walmart-paid education benefit program for full-time and part-time associates in Walmart and Sam's Club facilities. Programs range from high school completion to bachelor's degrees, including English Language Learning and short-form certificates. Tuition, books, and fees are completely paid for by Walmart.

Eligibility requirements apply to some benefits and may depend on your job classification and length of employment. Benefits are subject to change and may be subject to a specific plan or program terms.

For information about benefits and eligibility, see [One.Walmart](#).

The annual salary range for this position is \$143,000.00 - \$286,000.00

Additional compensation includes annual or quarterly performance bonuses.

Additional compensation for certain positions may also include :

- Stock

Minimum Qualifications...

Outlined below are the required minimum qualifications for this position. If none are listed, there are no minimum qualifications.

Option 1: Bachelor's degree in Computer Science and 5 years' experience in software engineering or related field.

Option 2: 7 years' experience in software engineering or related field. Option 3: Master's degree in Computer Science and 3 years' experience in software engineering or related field.

4 years' experience in data engineering, database engineering, business intelligence, or business analytics.

1 year's supervisory experience.

Preferred Qualifications...

Outlined below are the optional preferred qualifications for this position. If none are listed, there are no preferred qualifications.

Data engineering, database engineering, business intelligence, or business analytics, ETL tools and working with large data sets in the cloud, Master's degree in Computer Science or related field and 5 years' experience in software engineering or related field, We value candidates with a background in creating inclusive digital experiences, demonstrating knowledge in implementing Web Content Accessibility Guidelines (WCAG) 2.2 AA standards, assistive technologies, and integrating digital accessibility seamlessly. The ideal candidate would have knowledge of accessibility best practices and join us as we continue to create accessible products and services following Walmart's accessibility standards and guidelines for supporting an inclusive culture.

Primary Location...

809 11th Ave, Sunnyvale, CA 94089-4731, United States of America

Walmart and its subsidiaries are committed to maintaining a drug-free workplace and has a no tolerance policy regarding the use of illegal drugs and alcohol on the job. This policy applies to all employees and aims to create a safe and productive work environment.

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About Us



About Walmart:

Fifty years ago, Sam Walton started a single mom-and-pop shop and transformed it into the world's biggest retailer. Since those founding days, one thing has remained consistent: our commitment to helping our customers save money so they can live better. Today, we're reinventing the shopping experience and our associates are at

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